



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (EVT) (CBCS) Degree Examinations
 May – June 2025

22EVT222 – SOLAR BATTERY CHARGING SYSTEM

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define the following parameters of the solar cells, i. Short circuit current ii. Open circuit voltage iii. Maximum power point iv. Maximum power point current v. Maximum power point voltage vi. Fill factor vii. Solar cell efficiency	7	L2	1	3
	b) Explain the design procedure to estimate the number of cells in a PV module with the required diagram.	7	L2	1	3
	c) Compute the solar cell efficiency, output voltage, and maximum power when the cell operating temperature is 50 °C. It is given that the cell is a crystalline silicon cell, the cell efficiency and output voltage at standard test condition is 15% and 0.7 V respectively, the change in efficiency per degree Celsius rise in cell temperature from standard test condition value of 25°C is 0.6%/°C and the output voltage decrease per degree Celsius is 2.5 mV/°C. The maximum power at standard test condition is 3 W and decrease in maximum power per degree Celsius increase in cell temperature from standard test condition is 0.45%/°C.	6	L3	1	3
2.	a) Illustrate the current-voltage characteristics of the solar cell and explain the common factors that affect the power generated by solar cells.	8	L2	1	3
	b) Explain the procedure for plotting the current-voltage curve of the solar PV module using various equipment with the required diagram.	6	L2	2	3
	c) A solar cell having an area of 45 cm ² gives a current of 0.9 A and voltage of 0.7 V at maximum power point. The short circuit current is 0.95 A and open circuit voltage is 0.8 V. Input power is 1000 W/m ² . What is the Fill Factor, maximum power point and efficiency of the solar cell? Consider standard temperature condition.	6	L3	2	3
3.	a) Explain the design procedure for estimating the number of PV modules to be connected in series and the maximum power with the required diagram.	8	L2	2	3
	b) Determine the number of solar PV modules to be connected in parallel to achieve the current at peak power point of 42 A. The system voltage requirement is 16 volts. The modules to be connected are having the following parameters: maximum voltage $V_m = 16$ V, maximum current $I_m = 7$ A, open circuit voltage $V_{oc} = 20$ V, short circuit current $I_{sc} = 8.5$ A. Also, estimate the maximum power.	8	L4	2	3



- c) Three PV modules connected in series gives a maximum voltage V_m of 18 V and a maximum current I_m of 6 A.
- There is a mismatch in voltage of modules 2 and 3 with V_m of 17.7 V and 17.9 V. Compute, the total power produced by the series connected PV modules.
 - There is a mismatch in current of modules 2 and 3 with I_m of 6.1 A and 5.9 A. Compute, the total power produced by the series connected PV modules.

4 L3 2 3

Unit - II

4. a) Explain the commonly used set points and their behavior in the charge controller of solar PV system with the required diagram.
- b) For a DC to DC converter as shown in Figure Q4 (b), a battery of 32 V is connected and the input current is 5 A. At the output side, a voltmeter shows 60 V across the load and the output current is 2.5 A. Determine the following.

8 L2 3 3

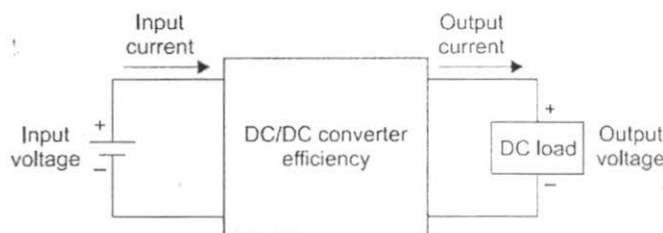


Figure: Q4 (b)

- Power input to the converter.
- Power output by the converter or absorbed by the load.
- Power lost in the converter due to losses.
- Efficiency of the converter.

6 L3 3 3

- c) Explain the operation of the boost converter in continuous conduction mode with the required diagram.

6 L2 3 3

5. a) Explain the maximum power point and sun tracking mechanism for high power extraction from solar photovoltaic systems with the required diagram.

6 L2 3 3

- b) A house has the following AC loads rated at 230 V.

- Four 70 W lights
- Three 100 W fans
- A 60 W radio

All the loads are powered simultaneously. A 24 V battery is available. Select the appropriate inverter specifications. Assume that the inverter efficiency is 90%.

8 L4 3 3

- c) Explain the concept and operation of a standalone photovoltaic system with the required diagram.

6 L2 4 3

6. a) Explain the basic components of solar street light systems. Also, Discuss their advantages and disadvantages.

7 L2 4 3

- b) Discuss the design procedure of solar photovoltaic system connected to a battery with the required diagram.

7 L2 4 3

- c) Explain the wire specifications required in designing SPV plants.

6 L2 4 3

Unit - III

7. a) Explain the electrical safety aspects for the installation of the solar photovoltaic systems.

6 L2 5 3

- b) Discuss the electrical testing procedure of the Inverter unit in solar photovoltaic systems.

8 L2 5 3

- | | | | | | |
|-------|--|---|----|---|---|
| c) | Explain the concept and operation of a grid connected photovoltaic system with the required diagram. | 6 | L2 | 5 | 3 |
| 8. a) | Explain the detailed procedure for the installation of solar photovoltaic array support structure. | 7 | L2 | 5 | 3 |
| b) | Discuss the testing procedure of the islanding protection in solar photovoltaic systems. | 7 | L2 | 5 | 3 |
| c) | Briefly explain the different types of grid Connected inverters. | 6 | L2 | 5 | 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
